The optimal constant in a non-tangential boundary inequality for convex bodies

Research note

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Abstract

For a convex body $K \subset \mathbb{R}^d$ and a point $x \in K$, let $(\partial K)_x$ be the boundary points at which the segment from x arrives non-tangentially. The divergence theorem gives

$$\mathcal{H}^{d-1}((\partial K)_x) \geq d \, \frac{|K|}{\operatorname{diam}(K)}.$$

The constant d is optimal if boundary base points and nonsmooth convex bodies are allowed: thin cones with x at the vertex approach equality. There is a wording issue in the stated open problem. If x is required to be an interior point of an open convex domain, then $(\partial K)_x = \partial K$ and the optimal constant is instead $2d$, by the classical isoperimetric and isodiametric inequalities. Thus the two natural readings both have exact answers.

1 The boundary-point version

Let $K \subset \mathbb{R}^d$ be a bounded convex body with nonempty interior, and let $x \in K$. At every regular point $y \in \partial K$, write $\nu(y)$ for the outward unit normal. Convexity makes the segment

$$\gamma(t) = x + t(y - x), \qquad 0 \leq t \leq 1,$$

a shortest path from x to y. The choice of constant-speed normalization does not affect whether the arrival is tangential. Accordingly, up to an $\mathcal{H}^{d-1}$-null set,

$$(\partial K)_x = \{y \in \partial K : \langle y - x, \nu(y) \rangle \neq 0\}.$$

Theorem 1. *For every bounded convex body $K \subset \mathbb{R}^d$ and every $x \in K$,*

$$\boxed{\mathcal{H}^{d-1}((\partial K)_x) \geq d \, \frac{|K|}{\operatorname{diam}(K)}.}$$

The constant d is best possible in the class of convex bodies when x may belong to the boundary.

Proof. At almost every $y \in \partial K$, the supporting-hyperplane inequality gives

$$\langle y - x, \nu(y) \rangle \geq 0.$$

Apply the divergence theorem to the vector field $F(z) = z - x$. Since $\operatorname{div} F = d$,

$$\begin{aligned} d|K| &= \int_{\partial K} \langle y - x, \nu(y) \rangle \, d\mathcal{H}^{d-1}(y) \\ &= \int_{(\partial K)_x} \langle y - x, \nu(y) \rangle \, d\mathcal{H}^{d-1}(y) \\ &\leq \operatorname{diam}(K) \, \mathcal{H}^{d-1}((\partial K)_x). \end{aligned}$$

The second line holds because the integrand vanishes precisely on the tangential part, modulo null sets, and the last line follows from $\langle y-x, \nu(y)\rangle \le |y-x| \le \operatorname{diam}(K)$.

It remains to verify sharpness. Let ω_{d-1} be the volume of the unit ball in $\mathbb{R}^{d-1}$ and, for $0<\varepsilon<1/\sqrt{3}$, consider the right circular cone

$$K_\varepsilon = \{(z,t) \in \mathbb{R}^{d-1} \times \mathbb{R} : 0 \le t \le 1,\ |z| \le \varepsilon t\}, \qquad x=(0,0).$$

The radial segment from the vertex is tangent to the lateral boundary. Apart from lower-dimensional edges, $(\partial K_\varepsilon)_x$ is exactly the base $\{t=1,\ |z|\le\varepsilon\}$. Hence

$$\mathcal{H}^{d-1}((\partial K_\varepsilon)_x) = \omega_{d-1}\varepsilon^{d-1}, \qquad |K_\varepsilon| = \frac{\omega_{d-1}\varepsilon^{d-1}}{d}.$$

For the stated range of ε, $\operatorname{diam}(K_\varepsilon)=\sqrt{1+\varepsilon^2}$. Therefore

$$\frac{\mathcal{H}^{d-1}((\partial K_\varepsilon)_x)\operatorname{diam}(K_\varepsilon)}{|K_\varepsilon|} = d\sqrt{1+\varepsilon^2} \longrightarrow d.$$

No larger universal constant is possible. □

The same calculation actually records a slightly stronger, point-dependent inequality. If

$$h_x = \operatorname*{ess\,sup}_{y\in(\partial K)_x} \langle y-x, \nu(y)\rangle,$$

then

$$\mathcal{H}^{d-1}((\partial K)_x) \ge \frac{d|K|}{h_x} \ge \frac{d|K|}{\operatorname{diam}(K)}. \tag{1}$$

For the cones above, the first inequality in (1) is an equality: the normal support distance is identically 1 on the base.

Remark 1. *The proof only needs that the Euclidean segment from x to almost every boundary point is the relevant geodesic. Thus the same argument applies, with the same constant, to bounded Lipschitz domains that are star-shaped with respect to x and whose radial description has the expected almost-everywhere normal behavior.*

2 Why the location of x changes the answer

The problem statement says both that the boundary is smooth and that $x\in\Omega$, language which ordinarily puts x in the interior. Its proposed near-extremizer, however, is a narrow sector with x at the corner, which is a boundary point and is not smooth there. These are genuinely different formulations.

Proposition 1. *If K is convex and $x \in \operatorname{int} K$, then $(\partial K)_x = \partial K$ up to a null set, and*

$$\mathcal{H}^{d-1}((\partial K)_x) \ge 2d\,\frac{|K|}{\operatorname{diam}(K)}.$$

The constant $2d$ is optimal, with equality for balls.

Proof. An interior point lies strictly below every supporting hyperplane, so $\langle y - x, \nu(y)\rangle > 0$ at every regular boundary point. Thus the non-tangential boundary is the full boundary. Write $D = \operatorname{diam}(K)$ and let ω_d denote the volume of the unit ball in $\mathbb{R}^d$. The isoperimetric inequality and the isodiametric inequality give, respectively,

$$\mathcal{H}^{d-1}(\partial K) \geq d\,\omega_d^{1/d}|K|^{(d-1)/d}, \qquad |K| \leq \omega_d (D/2)^d.$$

Combining them yields $\mathcal{H}^{d-1}(\partial K) \geq 2d|K|/D$. A ball attains equality in both classical inequalities. □

Consequently:

- for the intended convex-body formulation suggested by the sector example, allowing $x \in \partial K$, the answer is $c_d = d$;
- for the literal open-domain formulation $x \in \Omega = \operatorname{int} K$, the answer is $c_d = 2d$;
- if one separately insists that $x \in \partial K$ and that the boundary is globally C^1 (or smoother), Theorem 1 still supplies the lower bound d, but its cone sharpness example is outside that narrower class. That extra regularity variant should be stated as a separate question.

Novelty caveat

The argument is elementary enough that independent literature checking is especially appropriate. The motivating paper proves the constant $d - 1$ under a Laplacian condition and explicitly asks about the optimal constant; the January 2025 problem list still records that question. A targeted search did not locate the divergence-theorem observation above, but this note does not claim an exhaustive novelty search.

References

[1] S. Steinerberger, *The boundary of a graph and its isoperimetric inequality*, Discrete Applied Mathematics **338** (2023), 125–134; arXiv:2201.03489.

[2] S. Steinerberger, *Some Open Problems*, problem 30, version dated January 6, 2025, Xena Project problem-list page.